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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. Team & Project Overview

Team Name	Alt + Win + R
Project / Solution Name	Tepat Bill
Name (Leader)	Isaac Toh Zhen Yong
Email & Phone (Leader)	isaactoh2610@gmail.com & +60 10-557 6808
University Affiliation(s)	Universiti Malaya
Live Prototype URL	https://tepat-bill.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Srinivaas-only/Tepat-Bill-.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The mobile browser runs a React/TypeScript interface served with an Express API on Replit. Users upload document images or a PDF page and confirm extracted fields and received quantities. The backend stores document metadata and business records in PostgreSQL through Drizzle ORM; document bytes use Replit App Storage with a local-disk fallback.

Google Gemini is called server-side through Vercel AI SDK to extract structured fields. Zod validates the response. A hash cache, fallback model and manual-entry path handle repeated reads and failures. These paths do not constitute independent-reader voting.

Pure TypeScript rules compare items, quantities and reference prices and calculate RM differences in integer sen. Shared owner/staff PIN sessions and server permissions restrict approval actions. The owner reviews flags, opens an editable WhatsApp draft and records settlement outcomes. Supplier evidence uses a scoped public token link.

CSV export is a manual accounting handoff. Self-billed e-invoice release is a logged preview only. LHDN MyInvois submission code and a signed accounting webhook exist but are disabled unless credentials are configured; none are configured and none were used. The Orbyte link is a labelled simulation.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3; Node.js 20 (Replit module)	Application language and runtime	Apache-2.0; Node MIT and bundled notices
React/react-dom 19.3.1; Vite 8.3.5; Tailwind 4.3.3	Browser application and build	MIT
Express 5.3.0; pg 8.23.1; Drizzle ORM 0.45.4	API and PostgreSQL persistence	MIT; MIT; Apache-2.0

Name & Version	Purpose in Project	License
ai 7.0.149; @ai-sdk/google 4.0.103; Zod 4.6.7	Gemini calls and validation	Apache-2.0; Apache-2.0; MIT
@replit/object-storage 1.0.0; jose 6.2.13	Document storage and session tokens	MIT
pdfjs-dist 5.6.205; class-variance-authority 0.7.1	PDF first page; UI variants	Apache-2.0
@tanstack/react-query 5.104.5; motion 14.6.0; Radix UI (dialog 1.3.1, slot 1.5.0, tabs 1.2.1); sonner 2.0.8; clsx 2.1.1; tailwind-merge 3.7.0; tw-animate-css 1.4.0; react-zoom-pan-pinch 4.2.0; @number-flow/react 0.6.2; nanoid 5.1.16	UI, state, animation	MIT
wouter 3.13.0	Routing	Unlicense (public domain)
lucide-react 1.61.0	Icons	ISC
multer 2.5.0; cookie-parser 1.4.7; compression 1.8.2	Uploads, cookies, server utilities	MIT
Geist and Geist Mono (@fontsource-variable/* 5.3.0, self-hosted)	Fonts	SIL OFL-1.1
Kalam (Google Fonts)	Handwritten count font	SIL OFL-1.1
drizzle-kit 0.31.11; esbuild 0.28.2; tsx 4.23.15; vitest 5.0.5; supertest 7.3.1; fast-check 4.10.2; @vitejs/plugin-react; @tailwindcss/vite (development only)	Build and tests	MIT
Playwright (fixture generator only, not in package.json)	Renders the sample document images	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini Developer API	Server-side extraction of document images and structured fields. No fine-tuning.	Free tier
Replit App Storage SDK/service	Upload and retrieve document objects; local-disk fallback exists.	Sponsor-provided
WhatsApp wa.me deep link	Opens editable approved message and evidence link for human sending.	Free (consumer deep link, no API)

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit deployment	Hosts the web application and Express API; VM target in .replit.	Replit Core / North America
PostgreSQL	Replit-provisioned database (npg_9IWPEksmUdc6@ep-crimson-base-ary1rtkg.c-4.us-west-2.aws.neon.tech/neondb?sslmode=require)	Replit Core / North America

Service & Provider	Purpose in Project	Plan / Region

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gemini-3.8-flash	Google	pre-trained	Gemini API terms
gemini-3.5-flash-lite	Google	pre-trained	Gemini API terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic invoice fixtures	fixtures/docs/*.png and matching *.expected.json; fixtures/_generator/	Synthetic	Team-created; Apache-2.0	N
Seed data	fixtures/seed.json and synthetic test cases	Synthetic	Team-created; Apache-2.0	N
Department of Statistics Malaysia, Economic Census 2023: Wholesale and Retail Trade	https://www.dosm.gov.my/portal-main/release-content/economic-census-2023-wholesale-and-retail-trade-	Public	DOSM copyright — All rights reserved. Publicly available statistical publication	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Cafe-owner interview [22 October 2026]	-	Self-collected	Private, not published	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

[/] Yes

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code / Anthropic -Opus 5.5/ Sonnet 5.5	Refined the initial implementation afterward	Major
OpenAI Codex -6.1 Sol	Code review, browser checks, pitch and prototype consistency review and submission documents files	Major
Replit Agent; sonnet 5.5	Built the initial application	Major
Mobbin Pro via MCP	Referenced existing app screens and UI/UX patterns to guide the frontend design of Tepat Bill, including a Revolut home-screen reference	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We used Replit Agent to build the initial prototype and Claude Code to refine its implementation. Codex assisted code review, browser checks and documentation. Mobbin Pro via MCP provided frontend UI/UX references. Our team remains responsible for reviewing and adapting AI-generated output, checking calculations and behaviour, respecting third-party licences, and explaining the implementation. Automated tests are included in the repository, while automated tests in the repository cover the calculation, permission and evidence rules.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

[/] Yes

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Credits	Hosts the deployed prototype

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to the current submitted deployment because the application runs there. The Express/React/PostgreSQL architecture could be migrated to another host, but that would require configuring environment variables, persistent document storage and the database. Replit App Storage has a local-disk fallback in the code; this is not a verified production migration. Gemini is a separate document-reading dependency.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype stores uploaded documents, extracted supplier/item fields, receiving quantities, invoice amounts, flags, settlements and activity events. Demo fixtures are intended to be synthetic. Uploaded document bytes are sent to Google Gemini for extraction unless a cached or fixture result is used. Owner/manager PINs create signed HttpOnly session cookies; server permissions restrict business actions. Supplier evidence is accessible to anyone holding the scoped link, so forwarding can disclose its contents. Google Fonts receives normal browser requests.

The prototype has no automatic retention deadline, per-record deletion or evidence-link expiry. Demo reset does not establish complete erasure: extraction cache and stored document bytes can remain. Do not upload real personal or confidential data into this demo. This is not a claim of production privacy compliance.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Gemini may misread handwriting, item identity, quantities, prices or bounding boxes; provider outages and quota limits can prevent extraction. Schema validation, editable extraction review, fallback, caching and manual entry reduce disruption but do not guarantee correctness. Current extraction uses integer quantities and the browser PDF path renders the first page. Comparison relies on recorded counts and reference prices.

Independent review found inconsistent received counts, a count-to-invoice navigation failure, evidence-linking gaps and credit/currency edge cases. We fixed them and added automated regression tests (248 in the repository). We will re-test the deployed app on a phone before the demo. Shared role PINs provide limited individual accountability. Public evidence links have no time expiry, and data retention/deletion is incomplete. E-invoicing and Orbyte/accounting integration are previews or intended handoffs, not live statutory or financial services. No claim of perfect detection, guaranteed prevention of wrong payments or measured business savings is made.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

[/] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

[/] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

[/] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

[/] Any use of generative-AI tools for development has been disclosed and reviewed by the team.

[/] Any use of official SDC sponsor resources has been declared in Section 5.

[/] The prototype described here is accessible to judges through the links provided in Section 1.

[/] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[24/10/2026]